

Human Body and Nutrition (Part - 3)

Skeletal System

Human Body:

All organs work together to perform functions for the body. A group of organs doing some particular job for the body form an organ system.

Human body has the following system performing certain function:

Respiratory system is responsible for carrying oxygen from the air to the bloodstream. Oxygen is necessary for human life.

Digestive system breaks the food we eat into smaller substances. After digestion, food can be used for producing energy.

Skeletal system gives shape to human body and helps in the movement.

Nervous system performs the functions of thinking, smelling/seeing, tasting etc.

Circulatory system helps in transportation of nutrients and other substances to various parts of the body.

Muscular system helps in movement.

Skeletal System

It is the framework of bones which gives support to human body. Adult human skeleton has 206 bones. Different parts of the human skeleton are.

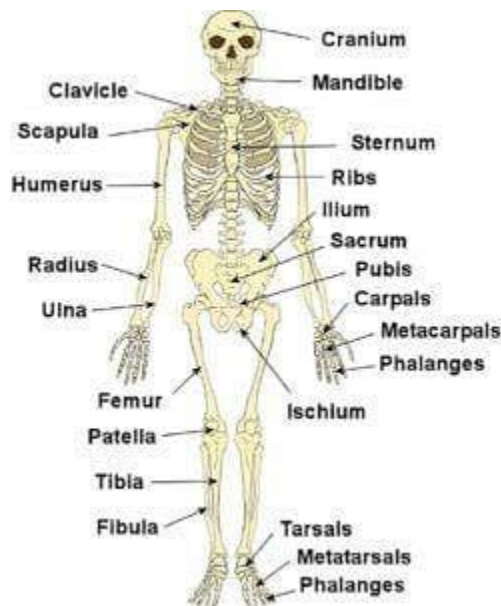


Fig: Skeleton System

Skull

Skull consists of 28 bones which protects the brain. All bones of skull are immovable

except the lower jaw. With the help of movable lower jaw, we can talk and eat. In lower and upper jaw, we have teeth for cutting and chewing food.

Rib Cage

Ribs make a cage of bones around the chest which is called rib cage. It protects our internal organs. Generally adults have 12 pairs of ribs. There is a long bone at the centre of the chest which holds the ribs in place which is called as sternum. Ribs are attached to the backbone. Last two ribs are not attached to the sternum and are known as floating ribs. These floating ribs are attached to the backbone.

Backbone

It protects the spinal cord. It is made up of a series of small bones called vertebrae. Backbone is also called as vertebral column. Tigers, frogs and some other animals have backbone and are called vertebrates. Snails, earthworm and cockroaches like animals do not have a backbone and are called invertebrates.

Limbs

All human beings have two pairs of limbs: the forelimbs (arms) and hind limbs (legs). Thigh bone or femur is the longest bone in the body.

Functions of the Skeletal System

The skeletal system has the following functions.

It gives shape and support to our body. Without the skeleton, our body would be floppy like a jelly. Forget about walking, we would not even be able to stand.

It protects our soft internal organs.

(i) The skull protects the brain.

(ii) The rib cage protects the heart and the lungs

(iii) The backbone protects the spinal cord.

It allows the movement of different body parts.

Bones contain marrow, where our blood cells are made.

Joints:

A Joint is the place where two bones meet. Most joints are movable. There are four kinds of joints in our body.

Ball and Socket Joint

This type of joint allows movement in many directions. The shoulder joint and the hip joint are examples of ball and socket joint.

Hinge joint

This type of joint works like the hinges in the door. This kind of joint only allows back and forth movement. Bones in the knee, elbow/ fingers, and toes have this type of joint.

Pivot joint

This type of joint is found between the first two vertebrae of your backbone. It allows you to move your head up, down, and sideways.

Gliding Joint

This type of joint is found in the bones of the wrist and the ankle. It allows these bones to slide against each other in a gliding motion.

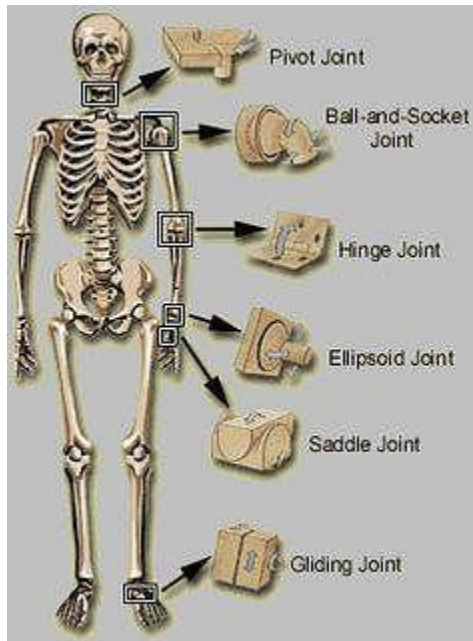


Fig: Different kinds of Joints in Human Body

Nervous System

The nervous system controls the different organs of our body.

Brain

Brain is a very important organ. It controls our body movements and helps us to store information in our memory. The human brain has three parts

Cerebrum

It is the largest part of the human brain and is responsible for learning, memory, intelligence, and logic.

Cerebellum

It is situated below the cerebrum and is responsible for muscle coordination and for maintaining the balance of our body.

Medulla or the Brain Stem

It controls activities such as heartbeat, breathing, swallowing, and sneezing.

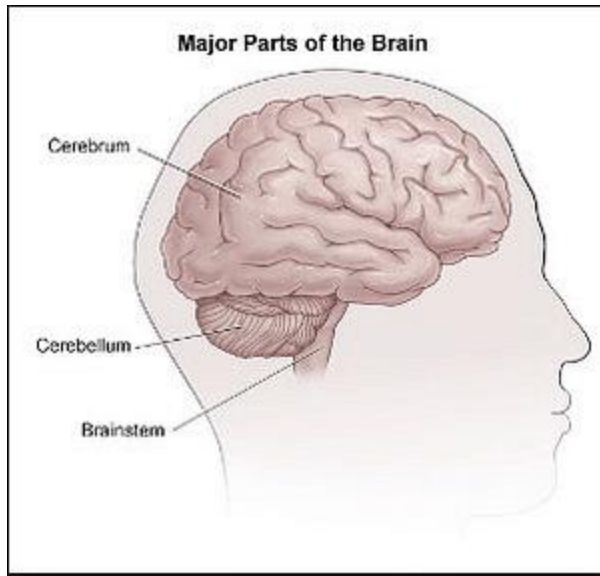


Fig: Different Organs of Nervous System

Spinal Cord

The spinal cord is a thick cord of nerve tissue, which extends from the brain stem. It is protected by the backbone. The spinal cord is responsible for the transfer of information between the brain and the rest of the body. It even controls the actions that do not involve the brain.

Nerves

A network of nerves runs throughout our body. Sensory nerves pass through the spinal cord and carry messages to the brain. Motor nerves carry messages back from the brain.

The automatic response of the body to an event is called a reflex action. Reflex actions are due to messages sent by the spinal cord. These actions are very fast. For example, if you touch a hot object accidentally you withdraw your hand almost immediately.

Sense Organs

Our sense organs help us to see, hear, taste, smell, and touch..

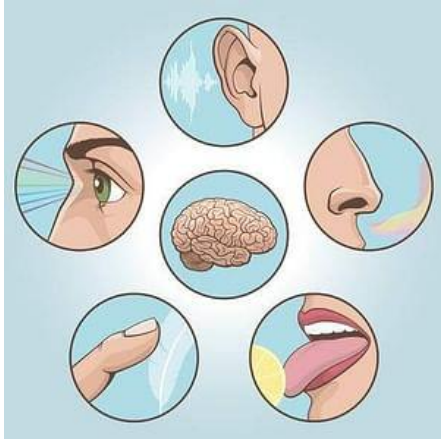


Fig: Different Senses of Human Body

Skin

There are tiny nerve endings in our skin. These nerve endings help us to feel things. They also help us to detect heat, cold, and pain.

Eyes

Eyes help us in seeing the various objects around us. Since eyes are very important organs, our body has several features to protect them. The eyebrows prevent sweat from running into the eyes. The eyelashes protect the eyes from dirt and strong light. The eyelids protect the eyes from injury.

Ears

Ears help us in hearing. They also help us to keep our balance.

Nose

Nose helps us to smell a variety of objects from flowers to rotten eggs.

Tongue

Tongue helps us in tasting things. Different regions of the tongue help us in tasting different kinds of substances.

Nutrition

Nutrition is defined as the process through which organisms take in and utilize materials necessary to support life. Food contains chemical substances called nutrients that are required by all living organisms. Our body requires carbohydrates, proteins, fats, minerals and vitamins. In addition to the five nutrients, our body also requires water and roughage.

Carbohydrates

Carbohydrates contain carbon, hydrogen and oxygen. Large percentage of food that we eat consists of carbohydrates. The simplest form of carbohydrate is sugar such as glucose and fructose. For example, glucose is present in food items like jams, jellies, etc. and fructose is present in various fruits such as banana, apple, etc.

Complex form of carbohydrate is starch. Starch is present in food items such as wheat, rice, potato, etc. Starch or sugar cannot be stored in the body.

Fats contain carbon, oxygen and hydrogen. Energy given by one gram of fat is twice the energy given by one gram of carbohydrate. Fat is present in food items such as butter, ghee, milk, grandaunts, etc. Our body takes long time to break down fat than to break down carbohydrate. Fats can be stored in the body. If the carbohydrates consumed are more than the required quantity, then excess amount is converted into fats and is stored in the body.

Proteins

Proteins contain nitrogen, carbon, oxygen and hydrogen. Proteins are required for the growth and repair of the body. Our hair, nails, skin and muscles are made up of proteins. Protein is found in food items such as pulses, soya bean, nuts, egg, meat, etc. Protein cannot be stored in our bodies. If excess amount of protein is consumed then it is excreted from the body in the form of urea or is stored as fats.